

UNIVERSITY OF MAKATI

College of Computing and Information Sciences
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PROJECT PROPOSAL

1st Year Java Programming with MySQL (JDBC)

Class Activity — Project Proposal Template

Sync Suites Hotel Reservation System

A desktop application that helps hotel staff manage room bookings, save customer details, process discounts, filter reservations, and check available rooms easily.

Group Name:	DEVSKADA
Leader:	Yabut, Renyer N.
Members:	Espijon, Arjay Lexiel, Gertos, Joshua Miguel Utrera, Sean John Daniel
Year & Section:	1st Year — EINS
Subject / Course:	Computer Programming 2
Instructor:	Ma. Corazon Benosa
Date Submitted:	May 17, 2026

GRADING RUBRIC

i Instructor Tip: This rubric is for instructor reference. Students should review this before writing their proposal.

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Work (1)
Project Description & Clarity	Very clear, complete	Mostly clear	Somewhat vague	Unclear/missing
Objectives (SMART)	All objectives SMART	Most are SMART	Some are SMART	Not SMART
Scope & Limitations	Well-defined both	Defined, minor gaps	Partially defined	Missing
Database Design (Tables/Fields)	Complete & correct ER	Minor errors	Incomplete tables	No DB design
JDBC / Tech Stack Description	Fully explained JDBC use	Mostly explained	Briefly mentioned	Not mentioned
Features & Modules	All modules detailed	Most described	Few described	Vague/missing
Team Roles & Timeline	Clear roles + timeline	Mostly clear	Partial only	Missing
Format & Completeness	Professional, complete	Minor format issues	Some sections missing	Incomplete
TOTAL SCORE	/32 points			

Scoring: 29–32 = Excellent | 21–28 = Good | 13–20 = Fair | Below 13 = Needs Revision

PART I — PROJECT OVERVIEW

1.1 Project Title

Sync Suites Hotel Reservation System

1.2 Project Description

Provide a brief but clear description of what your system does. Answer: What is the system? Who will use it? What problem does it solve?

This system helps hotel staff with their daily work. It allows them to add new customers or search for old ones. Staff can book rooms, automatically calculate prices (including extra charges and Senior/PWD discounts), and see which rooms are free in real-time. It uses a MySQL database to keep all the information safe. It also features advanced options to manage active guests, such as filtering booking records, extending stays, or upgrading room types.

1.3 Project Background / Problem Statement

Explain the current situation or problem that led to the need for this system.

Many small hotels still use paper logbooks to record guest details. This is very slow and causes mistakes, like booking the same room for two different people at the same time ("double booking"). Also, doing it manually makes it hard to track who has checked in, checked out, upgraded their rooms, or extended their stay. Searching for specific records is also a headache. Our system fixes these problems by making the process automatic, offering advanced filters to find records instantly, checking for errors, and showing the real status of every room.

1.4 Target Users

i Instructor Tip: Who will actually use and benefit from this system? Be specific.

Type of User	Role / How They Use the System
Administrator	Manages the system and logs in using a secure username and password.
Hotel Receptionist / Staff	Searches for customers, registers new guests, books rooms, applies discounts, checks guests in and out, upgrades rooms, extends stays, filters booking records, and looks at room availability.

PART II — OBJECTIVES

2.1 General Objective

State the main goal of your project in one to two sentences.

To design and build a simple Java desktop application connected to a MySQL database that manages hotel room bookings and tracks guest information easily. .

2.2 Specific Objectives

List at least four (4) specific, measurable objectives. Start each with an action verb (To develop, To implement, To create, To generate...).

i Instructor Tip: *Each objective should correspond to a major feature or module of your system.*

- 1. To build a Secure Login Module** so that only authorized hotel staff with a correct username and password can open the system.
- 2.To create a New Booking Module** that saves customer information. It will allow staff to quickly search for old customers to lock and auto-fill their details, set check-in/check-out dates, add guest counts, and automatically compute the final price with Senior/PWD discounts.
- 3. To make a Room Availability Module** that shows a clear list of all rooms with a Refresh button, automatically sorting them so "Occupied" and "Reserved" rooms are grouped at the top.
- 4. To build an Advanced Manage Booking Module** that lets staff update reservations. They can process Check-Ins, Check-Outs, Cancel bookings, Reschedule dates, Extend Stays, Upgrade Rooms, and easily filter booking records (by Status, Payment, or Room Type).

PART III — SCOPE AND LIMITATIONS

3.1 Scope (What the System Will Do)

List the specific functions, features, and modules that your system will include.

- **Admin login** using a username and password checked against the database.
- **Customer Records:** Add new customers or search for returning customers (locking their names to avoid accidental edits).
- **Bookings:** Book a room, set dates, add extra charges, apply Senior/PWD discounts, and compute the total price.
- **Room Status:** Show all room numbers and types in a table. Rooms are dynamically sorted (Occupied -> Reserved -> Available) and can be updated using a Refresh button.
- **Manage Reservations:** Allow staff to Check-In, Check-Out, and Cancel bookings.
- **Modifying Stays:** Staff can process an Upgrade Room (transferring a guest to a better room category), Extend Stay (adding more days to the check-out date), and Reschedule bookings (with an automatic 15% charge).
- **Filtering System:** Staff can filter the reservations table by *Status*, *Payment Method*, and *Room Type* using the Apply/Clear Filter buttons.
- **MySQL database connection** using JDBC.

3.2 Limitations (What the System Will NOT Do)

List the boundaries of your system — features or capabilities that are outside the scope of this project.

i Instructor Tip: *Be honest about limitations. This shows critical thinking, not weakness.*

- It is a desktop app only; it will not work on mobile phones or web browsers. .
- It does not process actual online payments. Staff must collect the payment manually and just record the payment method (Cash or GCash/QR Scan) in the system.
- The system will not send automatic emails or text messages to customers.
- It only works on a local computer (localhost), not across multiple hotel branches.

PART IV — SYSTEM FEATURES & MODULES

Describe each major module of your system. List the specific features/functions under each module.

i Instructor Tip: *Each module should match one or more of your specific objectives from Part II.*

#	Module Name	Features / Description
1	Login Module	Asks for a username and password. Supports pressing "Enter" to log in instantly. If correct, it opens the maximized main dashboard. If wrong, it shows a warning message. . .
2	New Booking Module	Lets staff fill out a form for a guest. It includes a search button to quickly find old customers. Staff can enter the guest count, select room types, check a box for Senior/PWD discounts, and see the total amount to pay.
3	Room Availability Module	A read-only table that lists every room. Features a "Refresh" button and automatically sorts the list based on status (Occupied first, then Reserved, then Available).
4	Manage Booking Module	A table of all reservations. Staff can click buttons to Check-In, Check-Out, or Cancel. It includes advanced buttons: Extend Stay, Upgrade Room, and Reschedule (adds a 15% charge). Features a Filter dialog to "Apply Filter" or "Clear Filter" based on Status, Payment, or Room Type.

PART V — DATABASE DESIGN

5.1 Database Overview

Provide the name of your MySQL database and a brief description of its structure.

Item	Details
Database Name	hotel_reservation
Number of Tables	5 tables
JDBC Connection URL	jdbc:mysql://localhost:3306/hotel_reservation";
MySQL Username	[e.g., root]
Java Driver Used	mysql-connector-j-9.6.0

5.2 Sample JDBC Connection Code

```
package FinalProject;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    private static final String URL      = "jdbc:mysql://localhost:3306/hotel";
    private static final String USERNAME = "root";    // default XAMPP username
    private static final String PASSWORD = "";        // default XAMPP password
    (empty)

    public static Connection getConnection() throws SQLException {
        try {
            // Load the MySQL JDBC driver (required before connecting)
            Class.forName("com.mysql.cj.jdbc.Driver");
            return DriverManager.getConnection(URL, USERNAME, PASSWORD);
        } catch (ClassNotFoundException e) {
            // This means the MySQL connector JAR is not in your project
            throw new SQLException("MySQL JDBC Driver not found. Add
mysql-connector-j.jar to your project !", e);
        }
    }
}
```

5.3 Database Tables

Define at least two (2) database tables your system will use. Add more tables as needed.

i Instructor Tip: Each table must have a Primary Key (PK). Use FK to indicate Foreign Keys linking to other tables.

Table 1: Admins - Stores the login details for the hotel staff or Admin

Field Name	Data Type	Constraint	Description
Admin_id	INT(11)	PK, AUTO_INCREMENT]	The unique number for each admin.
Username	[VARCHAR(50)]	[NOT NULL]	The login name used by the admin.
Password	[VARCHAR(100)]	[DEFAULT NULL]	The secret password for the admin.

Table 2: customers- Stores the personal information of the guests

Field Name	Data Type	Constraint	Description
customer_id	INT(11)	PK, AUTO_INCREMENT]	The unique number for each guest.
first_name	VARCHAR(50)	NOT NUL]	Stores The guest's first name.
last_name	VARCHAR(50)	NOT NULL	Stores The guest's last name.
phone_number	VARCHAR(30)	NOT NULL	Stores The Phone number of guest
email	VARCHAR(50)	NOT NULL	Stores The guest's email address.
created_at	DATETIME	Not Null, Default Current Timestamp	The exact date and time the guest was added to the system.

Table 3: room_types (Stores the different categories of rooms and their prices)

Field Name	Data Type	Constraint	Description
room_type_id	INT(11)	<i>PK, AUTO_INCREMENT</i>	The unique number for the room category.
type_name	VARCHAR(50)	NOT NULL]	The name of the room type (e.g., Standard Room, Junior Suite).
rate_per_day	DECIMAL(10,2)	NOT NULL	The price of the room per day (e.g., 2500.00).

Table 4: rooms (Stores the list of actual physical rooms in the hotel)

Field Name	Data Type	Constraint	Description
room_id	INT(11)	<i>PK, AUTO_INCREMENT]</i>	The unique number for the specific room.
room_number	VARCHAR(10)	NOT NULL	The door number of the room (e.g., '101', '205').
room_type_id	INT (11)	<i>Foreign Key</i>	Connects the room to its type and price.
is_available	TINYINT(1)	<i>Not Null, Default 1 (1 means available)</i>	Shows "1" if free, or "0" if someone is inside.

Table 5: bookings (Stores all the reservation transactions)

Field Name	Data Type	Constraint	Description
booking_id	INT(11)	<i>PK, AUTO_INCREMENT</i>	The unique transaction number.
customer_id	INT(11)	Foreign Key	Links to the guest making the reservation.
room_id	INT(11)	<i>Foreign Key (References rooms), Not Null</i>	Links to the specific room being rented.
check_in_date	DATE	Not Null	The day the guest will arrive.
check_out_date	DATE	Not Null	The day the guest will leave.
total_amount	DECIMAL(10,2)	Not Null	The final price they need to pay. .
payment_method	ENUM	Cash' or 'GCash/QR Scan', DEFAULT 'Cash'	How the guest paid Cash or Scan To pay
status	ENUM	Reserved', 'Checked In', 'Checked Out', 'Cancelled', DEFAULT 'Reserved'	Shows the exact current status of the stay. '.
reschedule_charge	DECIMAL(10,2)	Not Null, Default 0.00	Any extra fee added if the guest changes their dates.

PART VI — TECHNICAL SPECIFICATIONS

List the tools, technologies, and environment you will use to build the system.

Component	Specification / Details
Programming Language	Java SE (specify version, e.g., Java 21)
GUI / Interface	[Java Swing / JavaFX — specify which]
Database	MySQL [mysql-connector-j-9.6.0]
Database Connectivity	JDBC (Java Database Connectivity) — MySQL Connector/J
IDE / Editor	IntelliJ IDEA / Eclipse / VS Code
Operating System	Windows 11 / Linux
Database Tool	phpMyAdmin / XAMPP
Version Control	None
<i>[Other Tools]</i>	<i>[Specify any additional libraries or tools used]</i>

PART VII — TEAM ROLES & RESPONSIBILITIES

Assign a specific role and tasks to each group member. Every member must have a clear contribution.

i Instructor Tip: *Tip: Roles can overlap. All members should understand the full system, not just their part.*

Member Name	Role	Specific Responsibilities
Yabut, Renyer N.	Project Lead/Documentation, Backend developer,DB Admin	, Database design,, JDBC connection setup.Proposal document
Espijon, Arjay Lexiel	<i>Backend Developer/DB ADMIN</i>	CRUD operations in java, Database design, Debugging
Gertos, Joshua Miguel	Backend Developer/Frontend developer/ Documentation/Db Admin	Designing the frontend Gui and backend program and writting proposal document
Utrera,Sean John Daniel	QA / Documentation/front end	, System testing, bug reporting, user manual, proposal document,

PART VIII — PROJECT TIMELINE

Plan your project activities across 2 weeks. Be realistic about what can be accomplished each day.

i Instructor Tip: *Since this is a 2-week activity, focus on the most essential tasks. Divide work clearly among members.*

WEEK 1 — Planning, Database Design & Setup				
#	Day / Period	Activity / Task	Output	Assigned To
1	Day 1–2	Topic selection, group formation, role assignment	Approved topic & group roles	Yabut, Renyer N.
2	Day 2–3	<i>Finalize project title, description & objectives</i>	<i>Completed Parts I & II of proposal</i>	All Members
3	Day 3–4	Define scope, limitations, features & modules	Completed Parts III & IV	All Members
4	Day 4–5	<i>Design ER diagram and define all database tables</i>	<i>ER diagram & table definitions</i>	<i>Gertos, Joshua Miguel</i>
5	Day 5	Set up MySQL database and test JDBC connection in Java	Working DB connection (.java file)	Yabut, Renyer N.
6	Day 5–6	<i>Build basic GUI layout and navigation (Swing/JavaFX)</i>	<i>Main window with navigation</i>	<i>Utrera, Sean & Joshua Gertos</i>
7	Day 6–7	Complete and review full proposal document draft	Draft proposal document	All Members

WEEK 2 — Development, Testing & Final Submission				
#	Day / Period	Activity / Task	Output	Assigned To
8	Day 8–9	Implement Module 1: [Module Name] with JDBC CRUD	Working Module 1	Espijon, Arjay Lexiel
9	Day 9–10	<i>Implement Module 2: [Module Name] with JDBC CRUD</i>	<i>Working Module 2</i>	Espijon, Arjay Lexiel
10	Day 10–11	Implement Module 3: [Module Name] (if applicable)	Working Module 3	Espijon, Arjay Lexiel
11	Day 11	<i>Connect all modules — integration and navigation testing</i>	<i>Integrated system</i>	<i>All Members</i>
12	Day 12	System testing — identify and fix bugs	Bug-free or stable build	Utrera, Sean John Daniel & Yabut, Renyer N.
13	Day 12–13	<i>Finalize documentation, user manual & proposal</i>	<i>Complete proposal & user manual</i>	<i>All Members</i>
14	Day 13–14	Prepare presentation slides and rehearse demo	PowerPoint deck & demo ready	All Members
15	Day 14	<i>Final presentation and live system demonstration</i>	<i>Submitted project + demo done</i>	<i>All Members</i>

PART IX — EXPECTED OUTPUT

Describe what you will submit and deliver at the end of the project.

#	Deliverable	Description
1	Working Java Application (.jar / .exe)	Fully functional desktop app connected to MySQL via JDBC
2	MySQL Database Script (.sql file)	SQL file to create and populate the database
3	Source Code	Complete Java source code files organized by package
4	Project Proposal Document	This completed document with all sections filled in
5	ER Diagram	Entity-Relationship diagram showing all database tables and relations
6	User Manual	Step-by-step guide on how to install and use the system
7	Final Presentation (PowerPoint)	Slide deck used during the live demo and defense

PART X — CONCLUSION

Write a brief conclusion paragraph (3–5 sentences) summarizing your proposal.

i Instructor Tip: *Answer these in your conclusion: What is the system? Why is it important? What will the team gain from building it?*

The Sync Suites Hotel Reservation System is a practical Java desktop application that will automate room bookings using a MySQL database connected via JDBC. This system fixes the problem of slow and messy manual paper records by saving customer and booking details safely in a database. Through this project, our group will apply key programming concepts, learn how to connect Java to a database, and gain hands-on experience in building a real-world software solution.

PART XI — REFERENCES

List all references used in writing this proposal. Use APA format.

i Instructor Tip: Include at least 3 references: a Java textbook, MySQL documentation, and one online resource.

1. Oracle Corporation. (2024). Java SE Documentation. Retrieved from <https://docs.oracle.com/en/java/>
2. MySQL. (2024). MySQL 8.0 Reference Manual. Retrieved from <https://dev.mysql.com/doc/refman/8.0/en/>
3. Oracle Corporation. (2024). *Lesson: JDBC Basics*. The Java Tutorials. Retrieved from <https://docs.oracle.com/javase/tutorial/jdbc/basics/index.html>
4. Oracle Corporation. (2024). *Creating a GUI With Swing*. The Java Tutorials. Retrieved from <https://docs.oracle.com/javase/tutorial/uiswing/>
5. W3Schools. (2024). *SQL Tutorial*. Retrieved from <https://www.w3schools.com/sql/>

———— End of Project Proposal ————

[Group Leader Signature]
Group Leader

[Instructor Signature]
Instructor / Adviser